



GAC 135

QUANTITATIVE METHODS (GAC 135) UNIT 1

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LOGIC

1.1 Introduction

To understand mathematics and mathematical arguments, it is necessary to have a solid understanding of logic and the way in which known facts can be combined to prove new facts.

- The rules of logic give precise meaning to mathematical statements.
- The rules are used to distinguish between valid and invalid mathematical arguments.
- The language of mathematics consists primarily of declarative sentences. If a sentence can be classified as true or false, it is called a **statement**. The truth or falsity of a statement is known as its **truth value**.
- A proposition is any meaningful statement that is true or false, but not both.

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- Lowercase letters, such as $p, q, r, \dots$, would be used to represent propositions, but for compound propositions, uppercase letters, such as $P, Q, R, \dots$, would be used.
- The notation;

$$p : 1 + 1 = 3$$

would be used to define p as the proposition $1 + 1 = 3$.

- The truth value of a proposition is true, denoted by T , if it is a true statement and false, denoted by F , if it is a false statement.

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Example 1.1

Consider the following sentences.

- (a) Two plus two equals four.
- (b) Every continuous function is differentiable.
- (c) $x^2 - 5x + 6 = 0$.
- (d) A circle is the only convex set in the plane that has the same width in each direction.
- (e) Every even number greater than 2 is the sum of two primes.

Which of the sentences are **statements**?

Example 1.2

Which of the following sentences are statements.

- (a) If x is a real number, then $x^2 \geq 0$.
- (b) Seven is a prime number.
- (c) Seven is an even number.
- (d) This sentence is false.

1.2 PROPOSITIONAL LOGIC AND RELATED CONCEPTS

Propositions can be combined to form new sentences (compound sentences) using the connectives: **NOT, OR, AND, IF-THEN, and IF AND ONLY IF**.

Example 1.3

The Propositions:

- It is windy.
- The waves are high.

can be combined to form:

- It is not windy.
- It is windy or the waves are high.

- It is windy and the waves are high.
- If it is windy, then the waves are high.
- It is windy if and only if the waves are high.

1.3 Logical Connectives and Truth Tables

New propositions, called compound propositions or propositional functions, can be obtained from old ones (atomic proposition) p, q, r, s, t , by using symbolic connectives (sentential connectives or logical operations).

Negation ("NOT" connective)

Let p stand for a given statement. Then $\sim p$ (read not p) represents the logical opposite (negation) of p . When p is true, $\sim p$ is false; When p is false, $\sim p$ is true. This can be summarized in a truth table:

p	$\sim p$
T	F
F	T

where T stands for true and F stands for false.

Example 1.4

Let p, q , and r be given as follows:

p : Today is Monday, q : Five is an even number, r : The set of integers is countable.

Then their negations can be written as:

- $\sim p$: Today is not Monday.
- $\sim q$: Five is not an even number or, Five is an odd number.
- $\sim r$: The set of integers is not countable or, The set of integers is uncountable.

Note

Negation is one of the fundamental logical connectives used to construct compound propositions.

Conditional (Implication) ("IF, THEN" connective)

Let p and q be propositions. The implication $p \Rightarrow q$ is the proposition that is false only when p is true and q is false; otherwise, it is true. p is called the **hypothesis** (an antecedent), and q is called the **conclusion** (consequent).

It is important to recognize that in mathematical writing, the conditional statement can be disguised in several equivalent forms.

The following expressions all mean exactly the same thing:

If p , then q q provided that p
 p implies q q whenever p $p \rightarrow q$
 p only if q p is a sufficient condition for q
 q if p q is a necessary condition for p

$p \rightarrow q$
 p hypothesis (an antecedent)
 q conclusion (consequent)

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Disjunction ("OR" connective)

Let p and q be propositions.

The disjunction of p and q , denoted $p \vee q$, is the proposition: p or q . The "or" is used in an inclusive way. This proposition is false only when both p and q are false; otherwise, it is true.

Conjunction ("AND" connective)

The **conjunction** of p and q , denoted $p \wedge q$, is the proposition: p and q . This proposition is defined to be true only when both p and q are true; otherwise, it is false.

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Example 1.5

Identify the **antecedent** and the **consequent** in each of the following statements.

- (a) If n is an integer, then $2n$ is an even number.
- (b) You can work here only if you have a college degree.
- (c) The car will not run whenever you are out of gas.
- (d) Continuity is a necessary condition for differentiability.

$$\begin{array}{c|c|c|} \hline T & T & F \\ \hline F & F & F \\ \hline \end{array}$$

$p: n$ is an integer
 $q: 2n$ is an even number
 $p \rightarrow q$

Biconditional ("IF AND ONLY IF" connective)

The **biconditional** proposition of p and q , denoted by $p \leftrightarrow q$, is the propositional function that is true when both p and q have the same truth values and false if p and q have opposite truth values.

The connective $\leftrightarrow$ is called the **equivalence** connective.

Also reads, " p if and only if q " or " p is a necessary and sufficient condition for q ."

$$\begin{array}{c|c|c|} \hline p & q & p \leftrightarrow q \\ \hline T & T & T \\ T & F & F \\ F & T & F \\ F & F & T \\ \hline \end{array}$$

We found that the sentence

$$x^2 - 5x + 6 = 0$$

needed to be considered within a particular context in order to become a statement. When a sentence involves a variable such as x , it is customary to use functional notation when referring to it. Thus, we write

$$p(x) : x^2 - 5x + 6 = 0$$

to indicate that $p(x)$ is the sentence $x^2 - 5x + 6 = 0$. For a specific value of x , $p(x)$ becomes a statement that is either true or false. For example, $p(2)$ is true and $p(4)$ is false.

$$\begin{aligned} p(x) &: x^2 - 5x + 6 = 0 \\ p(2) &: 2^2 - 5(2) + 6 = 0 \\ &\quad p(2) \text{ true} \\ p(4) &: 4^2 - 5(4) + 6 = 2 \neq 0 \end{aligned}$$

1.6 PROPOSITIONS AND QUANTIFIERS

Statements such as $x > 3$ are often found in mathematical assertions. These statements are not propositions when the variables are not specified. However, one can produce propositions from such statements.

A **predicate** is an expression involving one or more variables defined on some domain, called the **domain of discourse**. Substitution of a particular value for the variable(s) produces a proposition, which is either true or false. For instance, $p(n) : n$ is prime is a predicate on the natural numbers. Observe that $p(1)$ is false, $p(2)$ is true. In the expression $p(x)$, x is called a free variable. As x varies, the truth value of $p(x)$ varies as well. The set of true values of a predicate $p(x)$ is called the truth set and will be denoted by T_p .

Within this context, we may remove the ambiguity of $p(x)$ by using a quantifier. The sentence
For every x , $x^2 - 5x + 6 = 0$
is a statement since it is false. In symbols, we write

$$\forall x, p(x),$$

where the **universal quantifier** $\forall$ is read, "For every...", "For all...", "For each...", or a similar equivalent phrase.

$$\forall x, p(x)$$

$$\text{For every } x, x^2 - 5x + 6 = 0 \text{ is false}$$

$$\exists x \exists x^2 - 5x + 6 = 0 \text{ True}$$

There exist

The sentence

There exist an x such that $x^2 - 5x + 6 = 0$
is also a statement, and it is true. In symbols, we write

$$\exists x \ni p(x),$$

where the **existential quantifier** $\exists$ is read, "There exists...", "There is at least one...", or something equivalent.

The symbol $\ni$ is just a shorthand notation for the phrase "such that."

$$\exists x \ni p(x)$$

such that

Or in the abbreviated form

$$\exists x < 7,$$

where it is understood that x represents a real number. Sometimes the quantifier is not explicitly written down, as in the statement:

If x is greater than 1, then x^2 is greater than 1.

The intended meaning is:

$$\forall x \text{ if } x > 1 \text{ then } x^2 > 1.$$

In general, if a variable is used in the antecedent of an implication without being explicitly quantified, the universal quantifier is assumed to apply.

Example 1.6

Rewrite each statement using $\exists$, $\forall$, and $\ni$, as appropriate.

- There exists a positive number x such that $x^2 = 5$.
- For every positive number M , there is a positive number N such that $N < \frac{1}{M}$.
- If $n \geq N$, then $|f_n(x) - f(x)| \leq 3$ for all x in A .
- No positive number x satisfies the equation $f(x) = 5$.

$$\forall x > 0, f(x) \neq 5$$

Some pencils are red
All chairs have four legs

No square matrices
are triangular

$$\exists x \ni$$

$$x > 0 \quad x \leq 0$$

$$x \geq 0 \quad x < 0$$

$$d) \sim [\exists x > 0 \ni x^2 = 5] \Leftrightarrow \forall x > 0, x^2 \neq 5$$

Negation

Symbolically, if

$p(x) : x$ is awake,

then

$$\sim [\forall x, p(x)] \iff [\exists x \sim p(x)]$$

where the symbol " $\sim$ " represents negation. Similarly,

$$\sim [\exists x \sim p(x)] \iff [\forall x \sim p(x)]$$

Example 1.7

Determine the truth value of each statement. Assume that x and y are real numbers. Justify your answers.

(a) $\forall x \exists y \ni x + y = 3$

(b) $\exists x \ni \forall y, x + y \neq 3$

Exercises

1. Mark each statement True or False. Justify each answer.

- (a) The symbol " $\forall$ " means "for every."
- (b) The negation of a universal statement is another universal statement.
- (c) The symbol " $\ni$ " is read "such that."

2. Mark each statement True or False. Justify each answer.

- (a) The symbol " $\exists$ " means "there exist several."
- (b) If a variable is used in the antecedent of an implication without being quantified, then the universal quantifier is assumed to apply.
- (c) The order in which quantifiers are used affects the truth value.

3. Write the negation of each statement.

- (a) Some pencils are red.
- (b) All chairs have four legs.
- (c) No one on the basketball team is over 6 feet 4 inches tall.
- (d) $\exists x > 2 \ni f(x) = 7$.
- (e) $\forall x$ in A , $\exists y > 2 \ni 0 < f(y) < f(x)$.
- (f) If $x > 3$, then $\exists \epsilon > 0 \ni x^2 > 9 + \epsilon$.

$$d) \sim [\exists x > 0 \ni x^2 = 5] \iff \forall x > 0, x^2 \neq 5$$

$$y) \sim [\forall m > 0, \exists n > 0 \ni n < \frac{1}{m}] \iff$$

$$\exists m > 0 \ni \forall n > 0, n \geq \frac{1}{m}$$

$$x < 0$$

4. Write the negation of each statement.

- (a) Everyone likes Robert.
- (b) Some students work part-time.
- (c) No square matrices are triangular.
- (d) $\exists x \text{ in } B \ni f(x) > k$.
- (e) If $x > 5$, then $f(x) < 3$ or $f(x) > 7$.
- (f) If $x \text{ in } A$, then $\exists y \text{ in } B \ni f(x) < f(y)$

5. Determine the truth value of each statement, assuming x is a real number. Justify your answer.

- (a) $\exists x \text{ in the interval } [2, 4] \ni x < 7$
- (b) $\forall x \text{ in the interval } \in [2, 4], x < 7$
- (c) $\exists x \ni x^2 = 5$
- (d) $\forall x, x^2 = 5$
- (e) $\exists x \ni x^2 \neq -3$
- (f) $\forall x, x^2 \neq -3$
- (g) $\exists x \ni x \div x = 1$
- (h) $\forall x, x \div x = 1$

6. Determine the truth value of each statement, assuming x is a real number. Justify your answer.

- (a) $\exists x \text{ in the interval } [3, 5] \ni x \geq 4$
- (b) $\forall x \text{ in the interval } [3, 5], x \geq 4$
- (c) $\exists x \ni x^2 \neq 3$
- (d) $\forall x, x^2 \neq 3$
- (e) $\exists x \ni x^2 = -5$
- (f) $\forall x, x^2 = -5$
- (g) $\exists x \ni x - x = 0$
- (h) $\forall x, x - x = 0$

7. Below are two strategies for determining the truth value of a statement involving a positive number x and another statement $P(x)$.

- (i) Find some $x > 0$ such that $P(x)$ is true.
- (ii) Let x be the name for any number greater than 0 and show $P(x)$ is true.

For each statement below, indicate which strategy is more appropriate.

- (a) $\forall x > 0, P(x)$
- (b) $\exists x > 0 \ni P(x)$
- (c) $\exists x > 0 \ni \sim P(x)$
- (d) $\forall x > 0, \sim P(x)$

8. Which of the following best identifies f as a constant function, where x and y are real numbers.

- (a) $\exists x \ni \forall y, f(x) = y$
- (b) $\forall x \exists y \ni f(x) = y$
- (c) $\exists y \ni \forall x, f(x) = y$
- (d) $\forall y \exists x \ni f(x) = y$

9. Determine the truth value of each statement, assuming that x and y are real numbers. Justify your answer. $\diamond$

- (a) $\forall x$ and $y, x \leq y$
- (b) $\exists x$ and y such that $x \leq y$
- (c) $\forall x, \exists y$ such that $x \leq y$
- (d) $\exists x$ such that $\forall y, x \leq y$

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10. Determine the truth value of each statement, assuming that x and y are real numbers. Justify your answer.

- (a) $\forall x, \exists y \ni xy = 0$
- (b) $\forall x, \exists y \ni xy = 1$
- (c) $\exists y \ni \forall x, xy = 1$
- (d) $\forall x, \exists y \ni xy = x$

11. Determine the truth value of each statement, assuming that x, y , and z are real numbers. Justify your answer.

- (a) $\exists x \ni \forall y, \exists z \ni x + y = z$
- (b) $\exists x$ and $y \ni \forall z, x + y = z$
- (c) $\forall x$ and $y, \exists z \ni y - z = x$
- (d) $\forall x$ and $y, \exists z \ni xz = y$
- (e) $\exists x \ni \forall y$ and $z, z > y \Rightarrow z > x + y$
- (f) $\forall x, \exists y$ and $z \ni z > y \Rightarrow z > x + y$